Scalar subspaces in rank-two Suzuki geometries

Human Compression project

Abstract

A maximal subfield need not be maximal as an invariant additive subspace. This distinction changes a family in the rank-two Suzuki classification. We give the exact overgroups in the relevant mixed parabolic, construct the resulting geometries, and determine their equivalence by field automorphisms. For Sz(32768) this single family has forty-one isomorphism classes in the source's convention, whereas its table gives fourteen classes in total. The argument reuses the scalar-center correspondence already established in this project's work on higher-rank Suzuki configurations. It does not classify the remaining families.

1 An elementary overgroup correspondence

We first state the reusable group-theoretic fact. All groups in this note are finite. An action is called primitive when it is transitive and has no nontrivial invariant partition.

Lemma 1. *Let V be an abelian normal subgroup of Q, let $H \le Q$, and suppose $Q = VH$. Put $W = V \cap H$. The overgroups of H in Q correspond bijectively to the H-invariant subgroups of V containing W, by*

$$L \longmapsto L \cap V, \qquad T \longmapsto TH.$$

The coset action on Q/H is primitive precisely when V/W has no proper nonzero H-invariant subgroup. It is doubly transitive precisely when H is transitive on the nonzero elements of V/W.

Proof. If $H \le L \le Q$, write an element of L as vh. Since $h \in L$, we have $v \in L \cap V$, and therefore $L = (L \cap V)H$. Conversely, an H-invariant subgroup $T \le V$ makes TH a subgroup; because $W \le T$, its intersection with V is T. These operations are inverse. The usual correspondence between blocks containing the base coset and overgroups of its stabilizer proves the assertion about primitivity. Finally, V/W acts regularly on the cosets, identified with its additive set, and the base stabilizer acts by conjugation. Its transitivity away from zero is exactly double transitivity. □

If V is a vector space over a field K and the image of H on V is the full group of nonzero scalar multiplications, the invariant additive subgroups are exactly the K-subspaces. Thus an overgroup TH covers H precisely when $\dim_K(T/W) = 1$. On those cosets its permutation group is the affine group of the line over K. This is the scalar-center argument from the earlier project manuscript; the statement above isolates the group-theoretic reason for it.

2 Application to the Suzuki root group

Let $n > d \geq 3$ be odd integers, with $d \mid n$, and put $F = \mathbb{F}_{2^n}$, $K = \mathbb{F}_{2^d}$, $q = 2^n$, $s = 2^d$ and $\theta = 2^{(n+1)/2}$. We use the standard faithful four-dimensional Suzuki matrices, with

$$U(a,b) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ a^{\theta+1} + b & a^\theta & 1 & 0 \\ a^{\theta+2} + ab + b^\theta & b & a & 1 \end{pmatrix}.$$

Multiplication gives the root-group law

$$U(a,b)U(c,e) = U(a+c, b+e+ac^\theta), \qquad Z_F = \{U(0,b) : b \in F\}.$$

The diagonal torus consists of

$$h(t) = \mathrm{diag}(t^{\theta+1}, t, t^{-1}, t^{-\theta-1}), \qquad h(t)^{-1}U(a,b)h(t) = U(t^\theta a, t^{\theta+2}b). \tag{1}$$

Let U_K be given by $a, b \in K$, let $H_K = \{h(t) : t \in K^*\}$, and put $B = U_K H_K$. These are the standard root group, torus and point stabilizer in $S = \mathrm{Sz}(s) \leq G = \mathrm{Sz}(q)$. The ambient exponent θ restricts to the Suzuki exponent on K: the two exponent indices differ by a multiple of d.

We retain the classical Suzuki facts that this matrix model is faithful, that B is the stabilizer in the doubly transitive action of S on $s^2 + 1$ ovoid points, that S is maximal in G when n/d is prime, and that $\mathrm{Aut}(G) = G \rtimes \langle \phi \rangle$ with $\phi : x \mapsto x^2$ of order n. These structural results are due to Suzuki and Tits, and are also the structural inputs to the source classification: see its Section 2.2 and the structural list in Section 4, which cite Suzuki's *On a class of doubly transitive groups* (Annals of Mathematics 75 (1962), 105–145), Tits and Lüneburg. We do not reprove these classical facts here.

Both exponents in (1) are relatively prime to $q - 1$. This is immediate for θ. For the other, an odd common divisor of $2^n - 1$ and $2^{(n-1)/2} + 1$ would give simultaneously $2^n = 1$ and $2^{n-1} = 1$ modulo that divisor, forcing it to be one. The action of H_K on Z_F is consequently the full group of K-scalar multiplications.

Theorem 2. *Put $Q = Z_F B$. The overgroups of B in Q are exactly*

$$Q_T = Z_T B \quad (K \leq T \leq F), \tag{2}$$

where T ranges over K-subspaces and $Z_T = \{U(0,t) : t \in T\}$. Their orders are $|Q_T| = |T|s(s-1)$. In particular, the groups covering B are exactly those with $\dim_K T = 2$, and their order is $s^3(s-1)$. For every $c \in F \setminus K$,

$$\langle B, U(0,c) \rangle = Z_{K+Kc} B. \tag{3}$$

Proof. Here Z_F is abelian and normal in Q, and $Z_F \cap B = Z_K$. The root group U_K centralizes Z_F; the diagonal action is the scalar action just proved. Lemma 1 gives (2), the order formula, and the maximality criterion. Conjugating $U(0,c)$ by H_K generates exactly the additive space Kc, and B already contains Z_K. This proves (3). □

The source's Lemma 6.9, printed page 112, asserts instead that (3) has order $qs(s-1)$ and contains B maximally because K is a maximal subfield. If $n/d = r$ is an odd prime, then $r \geq 3$. There are K-subspaces strictly between K and F, regardless of the absence of intermediate fields. Thus B is never maximal in this Q. This invalidates the corresponding mixed-parabolic row of Tables 8 and 11. The distinction is visible already at $s = 8$, $q = 512$:

$$|B| = 448 < 3584 < |Q| = 28672. \tag{4}$$

3 The actual geometries and their equivalence

Suppose from now on that $r = n/d$ is prime. For every two-dimensional K-subspace T containing K, form the rank-two coset geometry

$$\Gamma_T = \Gamma(G; S, Q_T). \tag{5}$$

As in the source, elements are left cosets and incidence means nonempty intersection. Isomorphism means an automorphism of the specified group G carrying one subgroup configuration to the other, with type interchange allowed. We make no assertion here about isomorphisms that do not preserve the specified group action.

Theorem 3. *Every geometry (5) is firm, residually connected, flag transitive, residually weakly primitive, and locally doubly transitive. Its two local degrees are $s^2 + 1$ and s. Two members are conjugate under G precisely when their spaces T are equal. Their isomorphism classes in the source's convention are exactly the orbits of ϕ on*

$$\mathbb{P}_K(F/K). \tag{6}$$

There are $(s^{r-1} - 1)/(s - 1)$ conjugacy classes before this field fusion.

Proof. Coordinates in the standard subfield matrices give $S \cap Q_T = B$. More explicitly, every element of Q_T has a unique expression $U(a, b)h(t)$ with $a \in K$, $b \in T$, $t \in K^*$; such an element is in S exactly when $b \in K$. The subgroup S is maximal in G. Since $T \neq K$, the group Q_T is not contained in S, and therefore $\langle S, Q_T \rangle = G$. The incidence graph is consequently connected. A rank-two coset system is a geometry and is flag transitive: every element has an incident partner, and each incident pair is a translate of the base pair. Its two valencies are $[S : B] = s^2 + 1$ and $[Q_T : B] = s$, so it is firm. In rank two, connectedness supplies all the residual connectivity conditions of rank at least two.

The stabilizer B is maximal in S by the classical primitive ovoid action, and is maximal in Q_T by Theorem 2. Along with maximality of S in G, these are exactly the source's rank-two residual weak primitivity conditions. The action of S on its local cosets is doubly transitive. By Lemma 1, the other local action is the affine action of $K \rtimes K^*$ on K, also doubly transitive. This proves the first assertions.

The group S is nonsoluble whereas Q_T is soluble, so an isomorphism cannot exchange their types. The joint normalizer of S and B in G is B: maximality and simplicity give $N_G(S) = S$, and B is self-normalizing in S because it fixes a unique ovoid point. Every element of B preserves each T, by the scalar action. Hence no two different spaces in (2) give conjugate configurations.

Both S and B are preserved by ϕ. Their joint normalizer in $\mathrm{Aut}(G)$ is thus $B \rtimes \langle \phi \rangle$. After the trivial B action is removed, equivalence is exactly field fusion of the spaces T. Such spaces are the inverse images of lines in F/K, giving (6) and its stated cardinality. □

4 Two exact field counts

The following counts use the full automorphism group of G, not just inner conjugacy. They require no Suzuki group census.

Lemma 4. *Let σ be an odd-order field automorphism of F preserving K. Every σ-fixed element of F/K has a representative in F^σ. Consequently*

$$(F/K)^\sigma = F^\sigma/K^\sigma.$$

Proof. If σ has odd order a and the class of v is fixed, the sum $v + \sigma(v) + \cdots + \sigma^{a-1}(v)$ is a fixed representative of that same class: its image in F/K is $av = v$ in characteristic two. The kernel of the map from fixed representatives is K^σ. □

First take $K = \mathbb{F}_8$, $F = \mathbb{F}_{512}$. There are nine lines in F/K. The map ϕ^3 is K-linear of order three. An invariant line would have an eigenvalue in K^* of order dividing three, hence eigenvalue one. But its fixed space is zero by Lemma 4, since $F^{\phi^3} = K$. Therefore ϕ^3 fixes no line. The orbits of the order-nine map ϕ can only have lengths one, three or nine; the first two would be fixed by ϕ^3. All nine lines form one orbit.

Now take $K = \mathbb{F}_8$, $F = \mathbb{F}_{32768}$. There are

$$(8^4 - 1)/7 = 585 \tag{7}$$

lines in F/K. The K-linear map ϕ^3 has order five and no invariant line: the same argument uses $\gcd(5, 7) = 1$ and $F^{\phi^3} = K$. Hence any power ϕ^j with $5 \nmid j$ fixes no line, because its generated group contains ϕ^3.

Put $\sigma = \phi^5$, of order three. Its fixed field is $\mathbb{F}_{32}$, and its fixed field on K is $\mathbb{F}_2$. Thus Lemma 4 gives a fixed subspace of F/K with sixteen elements. On K, σ acts as the fourth power. If a K-line is invariant, choose a nonzero vector v on it and write $\sigma(v) = \lambda v$. There is a unique $k \in K^*$ with $k^3 = \lambda^{-1}$, because $\gcd(3, 7) = 1$. Then kv is fixed. Conversely a fixed nonzero vector spans an invariant line. Each such line contains exactly one nonzero fixed vector, since the fixed scalar field is $\mathbb{F}_2$. There are therefore fifteen fixed lines for each of ϕ^5 and ϕ^{10}. Burnside's orbit count now gives

$$\frac{585 + 15 + 15}{15} = 41. \tag{8}$$

There are three orbits of length five and thirty-eight of length fifteen.

Corollary 5. *For $G = \mathrm{Sz}(32768)$, the geometries (5) with $S = \mathrm{Sz}(8)$ give forty-one isomorphism classes in the source's convention. All are locally doubly transitive. Theorem 7.1 and its Table 11 omit this family, and the total fourteen in Table 12 at $e = 7$ is false.*

Proof. The count is (8). The group Q_T has order 3584 and a nonabelian normal Sylow two-subgroup. None of the other rows with the nonsoluble parabolic Sz(8) has this subgroup type: their other parabolic is Suzuki, dihedral, a cyclic group extended by a group of order four, or the mixed group of order $32768 \cdot 8 \cdot 7$. Thus (5) is absent from the asserted exhaustive list. Forty-one classes in one family also contradict the printed total fourteen. The same order argument already gives an omitted family at $q = 512$. □

Attribution and scope. The target is D. Leemans, *The Rank 2 Geometries of the Simple Suzuki Groups* Sz(q), Beiträge zur Algebra und Geometrie 39(1) (1998), 97–120: Lemma 6.9, Theorem 7.1 and Tables 8, 11, 12. The source's classification is considered under its specified-action equivalence, defined on printed page 100. The matrix model, ovoid action, subgroup maximality and ambient automorphism description are retained classical inputs. The scalar-center argument was already proved and separately reviewed in the project's earlier Suzuki subfield manuscript, Section 4; the new contribution here is its application to this rank-two row, the genuine replacement family, its local double transitivity and its equivalence calculation. No historical priority is asserted. The remaining rows and the complete corrected global count are not established by this note. This corrects a published classification; it is not an answer to a named open conjecture.

The supplementary program independently constructs the fields of orders 512 and 32768, verifies their defining polynomials by Rabin's criterion, lists all two-dimensional $\mathbb{F}_8$-subspaces containing $\mathbb{F}_8$, and checks every field orbit. It agrees with (7)–(8). These finite controls are not premises of the written proof.